PRODUCT PROSPECTUS & MARKET ANALYSIS

Using Data to Drive Business Performance

Databases and Management Information Systems

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The Market Opportunity: Solving the "Two-Book Problem"

Currently, to teach a business-facing database course, instructors are forced to assign a **two-book bundle**, creating an inherent "stitching tax" on course design:

Book 1: Theory

Database Systems 14e (Coronel) is used to teach relational theory, normalization, and database concepts, but proves too abstract and engineering-focused for non-CS students.

Book 2: Execution

Illustrated Microsoft 365 1e (Beskeen) is used to teach students how to build forms and queries in Access, but functions as a software manual with no theoretical depth.

The Solution: *Using Data to Drive Business Performance* replaces this fragmented experience. It is a unified, single-textbook solution that integrates database theory directly with hands-on business execution, tailored specifically for the modern business undergraduate.

Head-to-Head Alignment Comparison

Feature	Database Systems 14e (Coronel)	Illustrated Microsoft 365 1e (Beskeen)	Using Data to Drive Business Performance
Primary Market Fit	CS and traditional IT/ MIS majors.	Vocational, entry-level software training.	**Business Analytics, Finance, Marketing, and modern MIS majors.**
Pedagogical Sequence	**Theory-First:** ERDs from scratch before writing queries.	**Click-First:** Menus, no structural rationale.	**The "Design Pivot": Query first (users), design second (creators).**
SQL & Tech Depth	Highly theoretical. Treats SQL as mathematical set theory.	Almost non-existent. Relies on Access QBE.	**Analytical SQL: Window functions, CTEs, and CASE logic for business questions.**
Cloud & AI	Bolted on. Relegated to end of book.	None. Strictly localized, desktop-bound.	**Native & Earned AI: Integrates Supabase, cloud RLS, and vector/RAG architecture.**
Business Strategy	Minimal. Focuses on system building.	None. Focuses on software competency.	**DIKW Pathway: Closes loop with BI reporting, KPI thresholds, decision memos.**

Key Differentiators & Pedagogical Innovation

1. The "Design Pivot"

Traditional texts front-load complex normalization (1NF-3NF) early on, demoralizing business students. This structure allows students to *use* data first—building confidence and context—before pivoting to *designing* schemas in Chapter 9.

2. Real SQL vs. QBE

While Access is a great starting point, modern employers demand real, executable SQL (the #1 requested data analyst skill). The text explicitly transitions students from local environments to modern dialects (PostgreSQL/SQLite).

3. "Earned AI" Architecture

Rather than slapping AI sidebars into old chapters, AI is taught as a structural consumer of databases—showing students that without clean schemas, AI hallucinations happen. It introduces modern vector/RAG concepts natively.

4. Major-Specific Career Translation

Chapter 16 explicitly maps database skills to specific business majors (Accounting, Finance, Marketing) and helps students translate lab work into resume-ready STAR interview answers.

5. AI-Resistant Assessment

Built structurally post-ChatGPT. End-of-chapter problems aren't simple lookup questions; they are artifact-based (ERDs, normalized schemas, decision memos) applied to an unseen scenario (PetVax), solving the assignment integrity problem at the architectural level.

6. Revision Economics & Modularity

Tool-dependent content (Access, Supabase, BI) is quarantined to specific modules. The relational spine (Ch. 2-10) is tool-neutral, allowing for surgical, cost-effective updates rather than full revision cycles.

Curriculum Alignment & Market Demand

- **IS2020 Competency Crosswalk:** The ACM/AIS IS2020 guidelines mandate Data and Information Management as a required competency area. This manuscript directly fulfills the core mandates.
- **Combating OER Affordability Pressures:** A cohesive, digital-first product provides the unified quality required to win seats back from free (but limited) OER texts into the publisher ecosystem.
- **A Tested, "Close-the-Loop" Architecture:** Built around a consistent six-file architecture. Every concept is taught theoretically and then immediately executed practically in the context of the running PetVax and Grading Database projects.